semantic leakage. The results account for all ten generations sampled from each model.

We find similar trends on LLAMA3-70B Instruct with an average human-annotated Leak-Rate of 66.7 and automatic evaluation of Leak-Rate that range from 71.2 to 77.3 across the different embedding types.

We also calculate Kendall's τ on the human annotations. We find high interannotator agreement between the human annotators ($\tau = 0.68$), indicating that humans generally agree on the cases that constitute semantic leakage. We also compute Kendall's τ on the human evaluation vs. similarity differences calculated using BERTScore embeddings to evaluate how well these methods correlate,[14] and get a moderate correlation of $\tau = 0.39$ when averaged between the two annotators.[15]

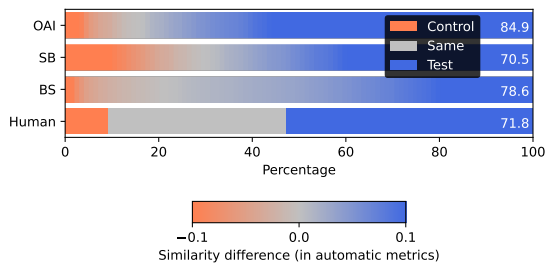


Figure 4: Human detection of semantic leakage compared to automatic methods. Leak-Rate is reported on the right for each method.

6 Multilingual and Crosslingual Semantic Leakage

In the previous sections we established that semantic leakage is exhibited in various scenarios in English, across different models and model sizes. We now inspect semantic leakage in languages outside of English, as well as in crosslingual settings, by focusing on Hebrew and Chinese, Hebrew-English and Chinese-English. As in the previous section, we consider the largest model from each model family at the temperature t found to leak the most: GPT4O ($t = 1$) and LLAMA3-70B Instruct ($t = 0$). Since we found that LLAMA is struggling with Hebrew, especially with Hebrew generation, we exclude these results.

[14]We introduce a slack variable of $\epsilon = 0.03$ to account for the different levels of similarity tolerance between humans and embedding-based similarity.

[15]While we use all model generations per example when obtaining label distributions and computing Leak-Rate, to calculate τ we use the generations shown to the human annotators, as this requires example-level alignments.

Multilingual Semantic Leakage We translate all the prompts into Chinese and Hebrew (see examples in Table 4 in the Appendix, rows 1, 3) with the help of native speakers (including the authors). The translation is mostly straightforward with the exception of idioms and names. For English idioms that do not exist in the target language, we write a new example using an idiom in that language (together with a matching context and a control prompt). For names, in Chinese we choose a plausible Chinese name with that meaning (e.g., 小麦, meaning *little wheat*, for Rye). In Hebrew, we make an attempt to translate to an existing frequent name in Hebrew, or to substitute it with a name with a related meaning (e.g., Lake instead of River). If that is not an option (e.g., Rye), we come up with a different name that has a clear semantic field.

Crosslingual Semantic Leakage We create mixed prompts that begin with Chinese/Hebrew and end with English (see examples in Table 4 in the Appendix, rows 2, 4). We use the translations to Chinese and Hebrew, and mix with English: for the first part of the prompt we use the translation to Chinese/Hebrew, and for the rest of it we use the original part in English. For evaluation purposes, we use the English concepts, since we expect the generation to be in English. For names or idioms that are used in Chinese/Hebrew parts of the prompt, we use their literal translations into English.

Results Figure 5 shows the human evaluation results for the multilingual and crosslingual experiments (with GPT4O). We can see significant semantic leakage in multilingual and crosslingual settings, with Leak-Rate values that range from 70.6 to 78.4 for the 4 different settings, similar to the Leak-Rate values we got for English.[16]

The generation quality with the LLAMA model is generally much lower, and we get Leak-Rates of 66.5 and 61.5 for Chinese and Chinese-English, respectively, according to human annotation. As noted above, the quality of generation in Hebrew and Hebrew-English did not allow for evaluation of semantic leakage.

Table 3 shows the automatic evaluation of the multilingual and crosslingual experiments. The results for the multilingual settings are not as reliable as in the English setting since we cannot determine

[16]In 6 examples of the crosslingual prompts (Zh-En), the generations are just translations because of the way the prompt is designed - we annotate those cases as "neutral".

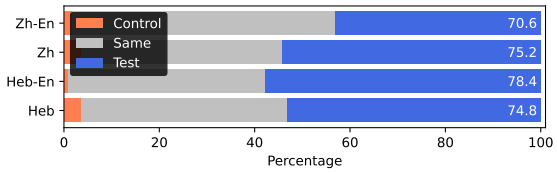


Figure 5: Human detection of semantic leakage in multilingual and crosslingual settings.

the quality of the underlying embedding methods for Hebrew and Chinese.[17] In the crosslingual setting we mainly compare English generations with English concepts, thus the results are more reliable.

Model	Language	Leak-Rate BS	SB	OAI
GPT4o	Heb	60.6	53.5	67.2
	Heb-En	62.1	58.2	74.9
	Zh	67.6	48.9	80.8
	Zh-En	61.6	60.5	71.3
3-70b-ins	Zh	73.4	54.1	82.8
	Zh-En	79.4	81.2	85.6

Table 3: Semantic leakage scores for multilingual and crosslingual setting, averaged across 10 samplings and measured by Leak-Rate.

7 Open-Ended Generation

We also examine semantic leakage in open-ended scenarios that encourage the model to generate multiple sentences or paragraphs. It is less clear how to quantify the semantic leakage in this setting, as it can manifest in more ways within the longer output; we therefore rely on more qualitative analysis in addition to the automatic metrics. This section focuses on generations from GPT4O with $t = 1$, as this model with this temperature was most prone to leakage in previous experiments (Section 4).

Stories A popular setting for open-ended text generation is storytelling: we ask the model to write a story about a child, with the child's name being the concept we expect to leak. Figure 6 gives the story that GPT4O generates when prompted to create a story about a child named Coral (and also Melody in Figure 9 in the Appendix). These generated stories contain many words and elements directly related to the semantics of each name, and the narrative of each story is tailored to those meanings. When measuring semantic leakage in this setting automatically for 23 different names,[18] we get Leak-Rates of 78.7, 60.9 and 76.1 with BS, SB and OAI embeddings respectively, suggesting similar rates of leakage to the other examples.

Semantic leakage in story generation

Prompt: *Tell me a short story about a child named Coral.*
Model: *Sure, here is a short story about a child named Coral: In the bustling coastal town of Seaview, ... and the waves danced upon the shore, ... Coral spent most of her days exploring the wonders of the seaside. ... She grabbed her seashell-collecting basket and marched down to the beach, ... Along the way, she encountered friendly crabs, playful dolphins ... telling stories of the ocean's many secrets. ... From that day on, Coral became known as the child who brought the ocean's magic to the town. She shared the shells, ... the mysteries of the sea ...*

Figure 6: Example of story generation, exhibiting semantic leakage based on the name in the prompt.

Recipes We also consider open-ended recipe generation. In Figure 7, we present GPT4O's generation when asked to suggest a recipe for use with a *blue* pan (or a *white* pan in Figure 10 in the Appendix), where we consider each color to be the potentially leaking concept. The model suggests to make blueberry pancakes and chicken alfredo, respectively, with the foods matching the color of the pan. When prompted with a red pan, ingredients like peppers, tomatoes and paprika are far more frequent, same as with asparagus in the case of a green pan and lemon with a yellow pan.

We experiment with 5 different colors (10 samples each) that are likely to be related to types of food (blue, green, red, white, and yellow). Automatic evaluation in this setting is more challenging since the concept might appear in the generation in a leaky way ("red pepper") or just as part of the answer, with no attributed leakage ("red pan") and these two cases are difficult to differentiate automatically. We decide to remove all occurrences of *"<color> pan"* from the generations when using automatic metrics, and get Leak-Rates

[17] To the best of our knowledge, the model underlying sentenceBERT was trained (predominantly) on English. This is why the metric detects more leakage in cross-lingual settings (with evaluations on English) compared to other languages.

[18] We remove any instance of the name from the generation, even when that instance is used as a common noun.

of 74.0/58.0/74.0 with BS/SB/OAI embeddings, again suggesting semantic leakage here as well.

Semantic leakage in recipe generation

Prompt: *I want to use my blue pan, give me a recipe:*
Model: *Sure! Here's a recipe for a delicious Blueberry Pancake that you can prepare using your blue pan: <recipe>*

Figure 7: Example of recipe generation, exhibiting semantic leakage based on the color in the prompt.

8 Related Work

Rassin et al. (2022) show similar conceptual leakage in image generation with DALLE-2 (Ramesh et al., 2022), a text-to-image model by OpenAI. They find that sense-ambiguous words are hard for the model to isolate—similar to our observation that idioms in the prompt can leak the literal meaning into the generation. Rassin et al. (2024) try to mitigate this issue by encouraging the cross-attention maps to agree with the linguistic binding reflected by the prompt syntax. Leakage of properties in DALLE-2 has also been discussed by Ramesh et al. (2022) and Leivada et al. (2023).

As models have improved, a related line of research has asked whether they match human behavior, particularly with respect to cognitive biases and priming. Prior work compares models' behavior to documented behavior of humans in psychological experiments, highlighting some similarities as well as differences from human response to given information (Jones and Steinhardt, 2022; Macmillan-Scott and Musolesi, 2024; Hagendorff et al., 2023). A known psychological phenomenon that is very related to models' semantic leakage as we present it here, is that of priming, and specifically semantic priming (Meyer and Schvaneveldt, 1971; Neely, 1976; Tulving et al., 1982; Tulving and Schacter, 1990), where decisions and reactions about specific words change according to given semantic stimuli provided before the target word. We plan to investigate this relationship further in future work.

Biases in Language Models There is an extensive line of work investigating bias and stereotypes of different demographic groups in NLP models, ranging from word embeddings (Bolukbasi et al., 2016; Caliskan et al., 2017; Gonen and Goldberg, 2019) all the way to LMs (Nadeem et al., 2021; Kotek et al., 2023; Cheng et al., 2023), sometimes focusing on specific end tasks (Gonen and Webster, 2020; Stanovsky et al., 2019; Kiritchenko and Mohammad, 2018; Zhao et al., 2018). Existing works in this space usually investigate specific bias types with respect to concrete socio-demographic groups (Kirk et al., 2021; Venkit et al., 2023, 2022; Navigli et al., 2023), as well as toxicity in model generation (Gehman et al., 2020; Nozza et al., 2021). Some works also point to specific input words that cause harmful generation, such as first names (De-Arteaga et al., 2019; Maudslay et al., 2019) among others (Röttger et al., 2024; Attanasio et al., 2022). In this context, semantic leakage serves as a broader class of association bias, potentially pointing to the underlying mechanism that causes biases in model generation by latching onto concrete properties in the input, resulting in toxic or stereotyped generation.

9 Conclusion

We identified a phenomenon in generative language models never discussed before, which we term *semantic leakage*: language models are prone to generating text that reflects unrelated semantic information from the prompt, leading to peculiar and sometimes unreasonable outputs. We defined and measured semantic leakage in a range of models, and together with human evaluation show that it is prevalent and consistent across all models we test. We also found that semantic leakage occurs in many different generation settings, including multilingual and crosslingual ones.

This leakage reflects associations learned by the model, similar to how different types of biases are learned; therefore, our characterization of semantic leakage broadens the scope we should consider with respect to potential ramifications of learned associations. While the implications of semantic leakage are not fully clear, it is a consistent behavior pattern in LMs that merits further study.

The finding that instruction-tuned models leak more is of special interest, given that they are the most popular and best performing model variations currently used. We hypothesize that semantic leakage is more dominant in these models because the leaking generations are less generic and seem to provide more information/content, which might be a property that is incentivized under these fine-tuning processes. We plan to explore this hypothesis more formally in future work.